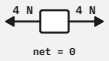
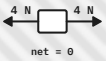
Say it, then write it: find the missing force

Language goal: I can **explain** what balanced means.



equilibrium
equilibrio · equilíbrio
a = 0, forces balance



balanced
equilibrado · equilibrado
net force = 0



support
soporte · apoio
a rope or surface that holds it up



missing force
fuerza faltante · força que falta
the one you must find

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: equilibrium · balanced · weight · up · zero

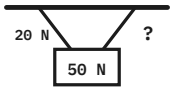
An object in ____ has zero acceleration. Its forces are ____. The upward forces must equal the ____. So the supports together pull ____ with the same size. The net force is ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The forces up add to ____ N and the forces down add to ____ N.
The missing force is ____ N because ____ .

3 · SOLVE WITH THE STEPS



A 50 N sign hangs from two ropes. Rope 1 pulls up 20 N.

STEP 1 · UP = DOWN
20 + ? = ____

STEP 2 · SUBTRACT
____ − ____ = ____ N

STEP 3 · SAY IT
"Rope 2 pulls up ____ N."

4 · EXPLAIN IT · CHOOSE A FRAME

□ The sign is in equilibrium because ____ .
□ Equilibrium does not mean stopped because ____ .
□ First I add the ____ forces, then I ____ .

I said it □ · I wrote it □

EQUILIBRIUM 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Three equilibrium myths are on the loose

18 Y

Fill each poster: the truth and evidence that proves it.

WANTED

FOR CRIMES AGAINST PHYSICS

"Equilibrium means the object is not moving."

LAST SEEN: a car on cruise control

THE TRUTH:

EVIDENCE:

REWARD: a = 0

WANTED

FOR CRIMES AGAINST PHYSICS

"If v = 0 for an instant, forces balance."

LAST SEEN: a ball at the top of a toss

THE TRUTH:

EVIDENCE:

REWARD: CHECK a

WANTED

FOR CRIMES AGAINST PHYSICS

"On a ramp, normal force equals weight."

LAST SEEN: a block resting on a ramp

THE TRUTH:

EVIDENCE:

REWARD: COMPONENTS

CAPTURED! · WRITE YOUR OWN WANTED NOTICE

A wrong idea about today's physics I once believed:

ANTOCCI ENGINEERING · DWG U1-18 · EQUILIBRIUM

SIGN HANGER: SCHOOL ENTRANCE

A 40 N sign hangs at rest. Design the ropes.

CALC 1 · TWO VERTICAL ROPES

Rope 1 holds 15 N. Rope 2 must hold ____ N.

CALC 2 · EQUAL ROPES

If both ropes share equally: ____ N each.

DRAWING · FBD OF THE SIGN (arrows to scale: 1 cm = 10 N)

CALC 3 · TILTED ROPES

Each upward part must be 20 N. Is the full rope tension more or less than 20 N? Why?

CALC 4 · SAFETY

Each rope is rated for 25 N. Which design is safe?

ENGINEER'S NOTE

State the condition that let you calculate: "Because the sign is ____, its acceleration is ____, so ____."

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Equilibrium and net force

18 H

DIRECTIONS · Fill all four boxes for each term.

Equilibrium

DEFINITION · MY WORDS

CONDITION

ΣF = ____ in every direction

EXAMPLES (2 kinds)

NON-EXAMPLES

Net force

DEFINITION · MY WORDS

HOW TO FIND IT

add forces with ____

SKETCH AN EXAMPLE

COMMON MIX-UP

One sentence using both terms: _____

EQUILIBRIUM 1 2 3

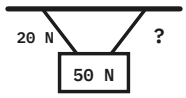
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Talk it through: balance

Language goal: I can **justify** a missing force.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · A 50 N SIGN, ROPE 1 = 20 N

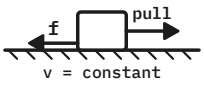


A ASKS "How much does rope 2 hold?"

B SAYS "The sign is at rest, so up = ____. Rope 2 = ____ − ____ = ____ N."

A CHECKS "So the upward forces add to ____."

ROUND 2 · SWITCH · A CRATE AT CONSTANT VELOCITY

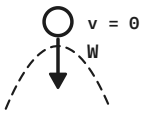


B ASKS "It's moving. Can the forces still balance?"

A SAYS "Yes. Constant velocity means a = ____, so ΣF = ____."

B ADDS "So friction must be ____ the pull."

ROUND 3 · BOTH · A BALL AT THE TOP OF A TOSS



A "Its velocity is ____ for an instant."

B "But it is NOT in equilibrium because ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

An object is in equilibrium when…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

EQUILIBRIUM 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: equilibrium

18 C

DIRECTIONS · Each object has zero acceleration. Copy each letter into the lock.

STATION 1

50 N sign, two vertical ropes. Rope 1 = 20 N. Rope 2?

C 30 N B 70 N

STATION 2

Crate at constant velocity; pull 12 N right. Friction?

A 12 N left E 0 N

STATION 3

15 N book on a level table. Normal force?

N 15 N up M 0 N

STATION 4

30 N lamp, two equal vertical ropes. Each rope?

C 15 N O 30 N

STATION 5

Ball at the very top of a toss. Equilibrium?

E no S yes

STATION 6

Car at a steady 25 m/s on a straight road. Net force?

L 0 N D 25 N

LOCK [1] [2] [3] [4] [5] [6]

Hint: in equilibrium, the forces ___ out.

EQUILIBRIUM 1 2 3

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Comic: not moving is not “no forces”

18 CS

YOUR READER
A younger student who thinks a still object has no forces on it.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **ORANGE** Rope (tension) arrows and the weight arrow
- ☐ ◌ **YELLOW** The word “equilibrium” in panel 3
- ☐ ◯ **PENCIL** “net force = 0 N”

1 · THE SIGN *A sign hanging from two ropes.*

CAPTION _____

2 · ALL THE FORCES *Draw every force on the sign.*

CAPTION _____

3 · THEY BALANCE *A character explains why the sign stays put.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **equilibrium · tension · weight · net force**

_____

_____

EQUILIBRIUM 1 2 3

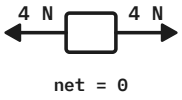
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Doodle Notes · equilibrium

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Equilibrium · equilibrio (ES) · equilíbrio (PT)



TODAY'S CASE

A cart moves straight at +2 m/s while equal and opposite horizontal forces act.

01 · SKETCH AND LABEL

Sketch its orange forces. Draw its blue position dots one second apart.

02 · CHANGE THE PICTURE

Show a second cart at rest with the same net force. Compare its position dots with those of the moving cart.

Annotate your sketches. Circle the mark that best shows equilibrium. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

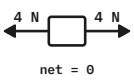
Both carts are in equilibrium because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture Dictionary · equilibrium

I can connect each word to a picture and use it to explain this case.

Name: _____ My home language: _____

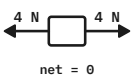


equilibrium

equilibrio · equilíbrio

a = 0, forces balance

My word or label: _____



balanced

equilibrado · equilibrado

net force = 0

My word or label: _____



support

soporte · apoio

a rope or surface that holds it up

My word or label: _____

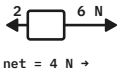


missing force

fuerza faltante · força que falta

the one you must find

My word or label: _____

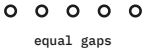


net force

fuerza neta · força resultante

the combined force on one object

My word or label: _____



constant velocity

velocidad constante · velocidade constante

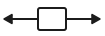
speed and direction both stay the same

My word or label: _____

MATCH · WRITE THE ENGLISH WORD UNDER EACH PICTURE



_____



_____



_____

Use two words to explain the case: A cart moves straight at +2 m/s while equal and opposite horizontal forces act.

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else